Quantum Mechanics I

Solutions to the Problems in the Final Exam

Fall, 2020

1. (25 points) Complete the following five problems (5 points each).

(1) In the $\{|x\rangle\}$ representation, show that the momentum operator $\hat{p}_x = -i\hbar\dfrac{d}{dx}$ in one dimension is a Hermitian operator.

Let $\psi(x)$ and $\varphi(x)$ be two arbitrary wave functions in one dimension with their values vanishing as $x \to \pm\infty$. We now consider the integral $\big(\psi, \hat{p}_x\varphi\big) = \displaystyle\int_{-\infty}^{\infty} dx\ \psi^*(x)\hat{p}_x\varphi(x)$. Making use of the representation of $\hat{p}_x$ in the $\{|x\rangle\}$ representation, $\hat{p}_x = -i\hbar\dfrac{d}{dx}$, we have

$$\big(\psi, \hat{p}_x\varphi\big) = -i\hbar\int_{-\infty}^{\infty} dx\ \psi^*(x)\frac{d\varphi(x)}{dx}.$$

Performing an integral by parts, we have

$$\big(\psi, \hat{p}_x\varphi\big) = -i\hbar\big[\,\psi^*(x)\varphi(x)\,\big]_{-\infty}^{\infty} + i\hbar\int_{-\infty}^{\infty} dx\ \frac{d\psi^*(x)}{dx}\varphi(x).$$

Making use of $\psi(x), \varphi(x) \to 0$ as $x \to \pm\infty$, we have

$$\big(\psi, \hat{p}_x\varphi\big) = i\hbar\int_{-\infty}^{\infty} dx\ \frac{d\psi^*(x)}{dx}\varphi(x).$$

Rewriting the right hand side of the above equation in terms of $\hat{p}_x$, we have

$$\big(\psi, \hat{p}_x\varphi\big) = \int_{-\infty}^{\infty} dx\ \left[-i\hbar\frac{d\psi(x)}{dx}\right]^*\varphi(x) = \int_{-\infty}^{\infty} dx\ \big[\hat{p}_x\psi(x)\big]^*\varphi(x) = \big(\hat{p}_x\psi, \varphi\big).$$

Comparing the above equation with the definition of the Hermitian conjugate of an operator, $\big(\hat{A}\psi, \varphi\big) = \big(\psi, \hat{A}^\dagger\varphi\big)$, we see that $\hat{p}_x^\dagger = \hat{p}_x$. That is, $\hat{p}_x$ is a Hermitian operator.

(2) Let $\varphi(x)$ be a stationary-state wave function obtained from the stationary Schrödinger equation $-\dfrac{\hbar^2}{2m}\dfrac{d^2\varphi(x)}{dx^2} + V(x)\varphi(x) = E\varphi(x)$ of a particle of mass m. Show that the first-order derivative $\dfrac{d\varphi(x)}{dx}$ of $\varphi(x)$ with respect to x is continuous at a finite potential jump located at x_0.

Integrating the stationary Schrödinger equation from $x_0 - \varepsilon$ to $x_0 + \varepsilon$ with $\varepsilon \to 0^+$, we obtain

$$\int_{x_0-\varepsilon}^{x_0+\varepsilon} dx\ \frac{d^2\varphi(x)}{dx^2} = \frac{2m}{\hbar^2}\int_{x_0-\varepsilon}^{x_0+\varepsilon} dx\ \big[\,V(x) - E\,\big]\varphi(x).$$

Performing the integral on the left hand side, we have

$$\frac{d\varphi(x)}{dx}\bigg|_{x_0-\varepsilon}^{x_0+\varepsilon} = \frac{2m}{\hbar^2}\int_{x_0-\varepsilon}^{x_0+\varepsilon} dx\ \big[\,V(x) - E\,\big]\varphi(x).$$

Since $V(x)$ is finite within the interval $[x_0 - \varepsilon, x_0 + \varepsilon]$ and $\varepsilon \to 0^+$, the value of the integral on the right hand side is zero. We then have

$$\frac{d\varphi(x)}{dx}\bigg|_{x_0-\varepsilon}^{x_0+\varepsilon} = 0.$$

Thus, the first-order derivative $\dfrac{d\varphi(x)}{dx}$ of $\varphi(x)$ with respect to x is continuous at a finite potential jump located at x_0.

(3) From the one-dimensional stationary Schrödinger equation of a particle of mass m in Dirac's notation, $\left[\frac{\hat{p}_x^2}{2m}+\hat{V}(\hat{x})\right]|\varphi\rangle = E\,|\varphi\rangle$, derive the one-dimensional stationary Schrödinger equation in momentum space.

Taking the inner product of $\left[\frac{\hat{p}_x^2}{2m}+\hat{V}(\hat{x})\right]|\varphi\rangle = E\,|\varphi\rangle$ with $|p_x\rangle$, we have

$$\langle p_x|\left[\frac{\hat{p}_x^2}{2m}+\hat{V}(\hat{x})\right]|\varphi\rangle = E\,\langle p_x|\varphi\rangle\,.$$

Making use of $\langle p_x|\varphi\rangle = \overline{\varphi}(p_x)$, $\langle p_x|\hat{p}_x^2|\varphi\rangle = p_x^2\overline{\varphi}(p_x)$, and $\langle p_x|\hat{V}(\hat{x})|\varphi\rangle = V\left(i\hbar\frac{d}{dp_x}\right)\overline{\varphi}(p_x)$, we have

$$\left[\frac{p_x^2}{2m}+V\left(i\hbar\frac{d}{dp_x}\right)\right]\overline{\varphi}(p_x) = E\overline{\varphi}(p_x).$$

The above equation is the one-dimensional stationary Schrödinger equation in momentum space.

(4) The propagator for the Schrödinger equation $i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = \hat{H}\big(\vec{r},-i\hbar\vec{\nabla}\big)\psi(\vec{r},t)$ is given by $K(\vec{r}_2,t_2;\vec{r}_1,t_1) = \theta(t_2-t_1)\sum_n e^{-i(t_2-t_1)E_n/\hbar}\varphi_n^*(\vec{r}_1)\varphi_n(\vec{r}_2)$, where E_n and $|\varphi_n\rangle$ are the eigenvalues and corresponding normalized eigenvectors of $\hat{H}$ that is time independent. Show that $\left[i\hbar\frac{\partial}{\partial t_2}-\hat{H}\big(\vec{r}_2,-i\hbar\vec{\nabla}_2\big)\right]K(\vec{r}_2,t_2;\vec{r}_1,t_1) = i\hbar\delta(t_2-t_1)\delta(\vec{r}_2-\vec{r}_1)$.

Applying $i\hbar\frac{\partial}{\partial t_2}-\hat{H}\big(\vec{r}_2,-i\hbar\vec{\nabla}_2\big)$ to $K(\vec{r}_2,t_2;\vec{r}_1,t_1)$ yields

$$\begin{aligned}
&\left[i\hbar\frac{\partial}{\partial t_2}-\hat{H}\big(\vec{r}_2,-i\hbar\vec{\nabla}_2\big)\right]K(\vec{r}_2,t_2;\vec{r}_1,t_1)\\
&\quad=\left[i\hbar\frac{\partial}{\partial t_2}-\hat{H}\big(\vec{r}_2,-i\hbar\vec{\nabla}_2\big)\right]\left[\theta(t_2-t_1)\sum_n e^{-iE_n(t_2-t_1)/\hbar}\varphi_n^*(\vec{r}_1)\varphi_n(\vec{r}_2)\right]\\
&\quad=i\hbar\delta(t_2-t_1)\sum_n e^{-i(t_2-t_1)E_n/\hbar}\varphi_n^*(\vec{r}_1)\varphi_n(\vec{r}_2)+\theta(t_2-t_1)\sum_n E_n e^{-i(t_2-t_1)E_n/\hbar}\varphi_n^*(\vec{r}_1)\varphi_n(\vec{r}_2)\\
&\qquad-\theta(t_2-t_1)\sum_n e^{-i(t_2-t_1)E_n/\hbar}\varphi_n^*(\vec{r}_1)\hat{H}\big(\vec{r}_2,-i\hbar\vec{\nabla}_2\big)\varphi_n(\vec{r}_2)\\
&\quad=i\hbar\delta(t_2-t_1)\sum_n e^{-iE_n(t_2-t_1)/\hbar}\varphi_n^*(\vec{r}_1)\varphi_n(\vec{r}_2)\\
&\quad=i\hbar\delta(t_2-t_1)\sum_n \varphi_n^*(\vec{r}_1)\varphi_n(\vec{r}_2)\\
&\quad=i\hbar\delta(t_2-t_1)\delta(\vec{r}_2-\vec{r}_1).
\end{aligned}$$

(5) Evaluate $\mathrm{Tr}\big(\sigma_\alpha\sigma_\beta\big)$ and $\mathrm{Tr}\big(\sigma_\alpha\sigma_\beta\sigma_\gamma\big)$, where $\alpha,\beta,\gamma = x,y,z$ and σ_x,σ_y, and σ_z are the Pauli matrices.
Formula provided: $\sigma_\alpha\sigma_\beta = \delta_{\alpha\beta}I + i\sum_\gamma \varepsilon_{\alpha\beta\gamma}\sigma_\gamma$.

Making use of $\sigma_\alpha\sigma_\beta = \delta_{\alpha\beta}I + i\sum_\gamma \varepsilon_{\alpha\beta\gamma}\sigma_\gamma$, we have

$$\mathrm{Tr}\big(\sigma_\alpha\sigma_\beta\big) = \delta_{\alpha\beta}\,\mathrm{Tr}\,I + i\sum_\gamma \varepsilon_{\alpha\beta\gamma}\,\mathrm{Tr}\,\sigma_\gamma.$$

Making use of $\mathrm{Tr}\,I = 2$ and $\mathrm{Tr}\,\sigma_\gamma = 0$, we have

$$\mathrm{Tr}\big(\sigma_\alpha\sigma_\beta\big) = 2\delta_{\alpha\beta}.$$

Multiplying $\sigma_\alpha\sigma_\beta = \delta_{\alpha\beta}I + i\sum_\rho \varepsilon_{\alpha\beta\rho}\sigma_\gamma$ with σ_γ from right yields

$$\sigma_\alpha\sigma_\beta\sigma_\gamma = \delta_{\alpha\beta}I\sigma_\gamma + i\sum_\rho \varepsilon_{\alpha\beta\rho}\sigma_\rho\sigma_\gamma = \delta_{\alpha\beta}\sigma_\gamma + i\sum_\rho \varepsilon_{\alpha\beta\rho}\sigma_\rho\sigma_\gamma.$$

Taking the trace of both sides of the above equation, we have

$$\mathrm{Tr}\big(\sigma_\alpha\sigma_\beta\sigma_\gamma\big) = \delta_{\alpha\beta}\,\mathrm{Tr}\,\sigma_\gamma + i\sum_\rho \varepsilon_{\alpha\beta\rho}\,\mathrm{Tr}\big(\sigma_\rho\sigma_\gamma\big).$$

Making use of $\mathrm{Tr}\,\sigma_\gamma = 0$ and $\mathrm{Tr}\big(\sigma_\rho\sigma_\gamma\big) = 2\delta_{\rho\gamma}$, we have

$$\mathrm{Tr}\big(\sigma_\alpha\sigma_\beta\sigma_\gamma\big) = 2i\sum_\rho \varepsilon_{\alpha\beta\rho}\delta_{\rho\gamma} = 2i\varepsilon_{\alpha\beta\gamma}.$$

2. (15 points) The Hamiltonian of a quantum system is given by $\hat{H} = \dfrac{\hat{\vec{p}}^2}{2m} + \hat{V}(\hat{\vec{r}})$ where $\hat{V}(\hat{\vec{r}})$ is the Hermitian potential energy operator. The eigenequation of $\hat{H}$ reads $\hat{H}\psi_n = E_n\psi_n$. Assume that the eigenvalue spectrum of $\hat{H}$ is discrete and that ψ_n's are normalized. Take $\hbar$ to be the parameter in the Hellmann-Feynman theorem.

(1) (5 points) Apply the Hellmann-Feynman theorem in real space.

(2) (5 points) Apply the Hellmann-Feynman theorem in momentum space.

(3) (5 points) Using the results obtained in the previous two parts, derive the virial theorem $\left(\psi_n, \dfrac{\hat{\vec{p}}^2}{2m}\psi_n\right) = \dfrac{1}{2}\big(\psi_n, \vec{r}\cdot\vec{\nabla}V(\vec{r})\psi_n\big)$; also written as $\big\langle\hat{T}\big\rangle_n = \dfrac{1}{2}\big\langle\vec{r}\cdot\vec{\nabla}V(\vec{r})\big\rangle_n$ with $\hat{T} = \dfrac{\hat{\vec{p}}^2}{2m}$ the kinetic energy operator.

Formula provided: $\dfrac{\partial E_n(\lambda)}{\partial\lambda} = \left(\psi_n(\lambda), \dfrac{\partial\hat{H}(\lambda)}{\partial\lambda}\psi_n(\lambda)\right)$.

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(1) (5 points) In real space, the Hamiltonian is given by

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r}).$$

The partial derivative of $\hat{H}$ with respect to $\hbar$ is given by

$$\frac{\partial\hat{H}}{\partial\hbar} = -\frac{\hbar}{m}\nabla^2.$$

Applying the Hellmann-Feynman theorem in real space, we obtain

$$\frac{\partial E_n}{\partial\hbar} = \left(\psi_n, \frac{\partial\hat{H}}{\partial\hbar}\psi_n\right) = \left(\psi_n, -\frac{\hbar}{m}\nabla^2\psi_n\right) = \frac{2}{\hbar}\left(\psi_n, \frac{\hat{\vec{p}}^2}{2m}\psi_n\right).$$

(2) (5 points) In momentum space, the Hamiltonian is given by

$$\hat{H} = \frac{\vec{p}^2}{2m} + V(i\hbar\vec{\nabla}_{\vec{p}}).$$

The partial derivative of $\hat{H}$ with respect to $\hbar$ is given by

$$\frac{\partial\hat{H}}{\partial\hbar} = \frac{1}{\hbar}\big[\vec{r}\cdot\vec{\nabla}V(\vec{r})\big]_{\vec{r}\to i\hbar\vec{\nabla}_{\vec{p}}}.$$

Let $\varphi_n(\vec{p})$ be the eigenfunction corresponding to E_n in momentum space. Applying the Hellmann-Feynman theorem in momentum space, we obtain

$$\frac{\partial E_n}{\partial\hbar} = \left(\varphi_n, \frac{\partial\hat{H}}{\partial\hbar}\varphi_n\right) = \frac{1}{\hbar}\left(\varphi_n, \big[\vec{r}\cdot\vec{\nabla}V(\vec{r})\big]_{\vec{r}\to i\hbar\vec{\nabla}_{\vec{p}}}\varphi_n\right).$$

The average on the right hand side can be also evaluated in real space. We have

$$\frac{\partial E_n}{\partial\hbar} = \frac{1}{\hbar}\big(\psi_n, \vec{r}\cdot\vec{\nabla}V(\vec{r})\psi_n\big).$$

(3) (5 points) From the above two results obtained respectively from applying the Hellmann-Feynman theorem in real and momentum spaces, we have

$$\left(\psi_n, \frac{\hat{\vec{p}}^2}{2m}\psi_n\right) = \frac{1}{2}\big(\psi_n, \vec{r}\cdot\vec{\nabla}V(\vec{r})\psi_n\big).$$

Making use of $\hat{T} = \dfrac{\hat{\vec{p}}^2}{2m}$ and the angular brackets for averages, we can put the above equation into the following form

$$\big\langle\hat{T}\big\rangle_n = \frac{1}{2}\big\langle\vec{r}\cdot\vec{\nabla}V(\vec{r})\big\rangle_n.$$

Multiplying the factor of 2 on both sides of the above equation, we can reduce the vertical space occupied by the equation

$$2\big\langle\hat{T}\big\rangle_n = \big\langle\vec{r}\cdot\vec{\nabla}V(\vec{r})\big\rangle_n$$

which is the mathematical expression of the virial theorem.

3. (15 points) A three-dimensional rotor is in the eigenstate $|\ell m\rangle$ common to $\hat{\vec{L}}^2$ and $\hat{L}_z$ with the state wave function in the $\{|\vec{r}\rangle\}$ representation given by $\langle\vec{r}|\ell m\rangle = Y_{\ell m}(\theta,\phi)$. Here $Y_{\ell m}(\theta,\phi)$ is a spherical harmonic function.

(1) (4 points) Find the averages $\langle\hat{L}_x\rangle$ and $\langle\hat{L}_y\rangle$ of $\hat{L}_x$ and $\hat{L}_y$ in $|\ell m\rangle$.

(2) (8 points) Find the uncertainties ΔL_x and ΔL_y of $\hat{L}_x$ and $\hat{L}_y$ in $|\ell m\rangle$.

(3) (3 points) Verify that the uncertainty relation for $\hat{L}_x$ and $\hat{L}_y$ is satisfied.

Formulas provided: $[\hat{L}_x,\hat{L}_y] = i\hbar\hat{L}_z,\ [\hat{L}_y,\hat{L}_z] = i\hbar\hat{L}_x,\ [\hat{L}_z,\hat{L}_x] = i\hbar\hat{L}_y,\ \hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y,$
$\hat{L}_\pm\,|\ell m\rangle = \hbar\sqrt{\ell(\ell+1) - m(m\pm1)}\,|\ell, m\pm1\rangle,\ \Delta A\Delta B \geq \dfrac{1}{2}\big|\langle[\hat{A},\hat{B}]\rangle\big|.$

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(1) (4 points) Making use of $[\hat{L}_y,\hat{L}_z] = i\hbar\hat{L}_x$, we have

$$\begin{aligned}\langle\hat{L}_x\rangle &= \frac{1}{i\hbar}\,\langle\ell m|[\hat{L}_y,\hat{L}_z]|\ell m\rangle = \frac{1}{i\hbar}\,\langle\ell m|\hat{L}_y\hat{L}_z - \hat{L}_z\hat{L}_y|\ell m\rangle \\ &= -im\big[\,\langle\ell m|\hat{L}_y|\ell m\rangle - \langle\ell m|\hat{L}_y|\ell m\rangle\,\big] = 0.\end{aligned}$$

Making use of $[\hat{L}_z,\hat{L}_x] = i\hbar\hat{L}_y$, we have

$$\begin{aligned}\langle\hat{L}_y\rangle &= \frac{1}{i\hbar}\,\langle\ell m|[\hat{L}_z,\hat{L}_x]|\ell m\rangle = \frac{1}{i\hbar}\,\langle\ell m|\hat{L}_z\hat{L}_x - \hat{L}_x\hat{L}_z|\ell m\rangle \\ &= -im\big[\,\langle\ell m|\hat{L}_x|\ell m\rangle - \langle\ell m|\hat{L}_x|\ell m\rangle\,\big] = 0.\end{aligned}$$

(2) (8 points) Expressing $\hat{L}_x$ and $\hat{L}_y$ in terms of $\hat{L}_\pm$, we have

$$\hat{L}_x = \frac{1}{2}\big(\hat{L}_+ + \hat{L}_-\big),\ \hat{L}_y = \frac{1}{2i}\big(\hat{L}_+ - \hat{L}_-\big).$$

Squaring the above two equations yields

$$\begin{aligned}\hat{L}_x^2 &= \frac{1}{4}\big(\hat{L}_+^2 + \hat{L}_-^2 + \hat{L}_+\hat{L}_- + \hat{L}_-\hat{L}_+\big), \\ \hat{L}_y^2 &= -\frac{1}{4}\big(\hat{L}_+^2 + \hat{L}_-^2 - \hat{L}_+\hat{L}_- - \hat{L}_-\hat{L}_+\big).\end{aligned}$$

Making use of $\langle\ell m|\hat{L}_\pm^2|\ell m\rangle = 0$, we have

$$\langle\hat{L}_x^2\rangle = \langle\hat{L}_y^2\rangle = \frac{1}{4}\,\langle\ell m|\hat{L}_+\hat{L}_- + \hat{L}_-\hat{L}_+|\ell m\rangle\,.$$

Making use of $\hat{\vec{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = \frac{1}{2}\big(\hat{L}_+\hat{L}_- + \hat{L}_-\hat{L}_+\big) + \hat{L}_z^2$, we have

$$\frac{1}{2}\big(\hat{L}_+\hat{L}_- + \hat{L}_-\hat{L}_+\big) = \hat{\vec{L}}^2 - \hat{L}_z^2.$$

We can then evaluate $\langle\hat{L}_x^2\rangle = \langle\hat{L}_y^2\rangle$ as follows

$$\langle\hat{L}_x^2\rangle = \langle\hat{L}_y^2\rangle = \frac{1}{2}\,\langle\ell m|\hat{\vec{L}}^2 - \hat{L}_z^2|\ell m\rangle = \frac{\hbar^2}{2}\big[\,\ell(\ell+1) - m^2\,\big].$$

The uncertainty in $\hat{L}_x$ is given by

$$\Delta L_x = \sqrt{\langle\hat{L}_x^2\rangle - \langle\hat{L}_x\rangle^2} = \frac{\hbar}{\sqrt{2}}\sqrt{\ell(\ell+1) - m^2}.$$

The uncertainty in $\hat{L}_y$ is given by

$$\Delta L_y = \sqrt{\langle\hat{L}_y^2\rangle - \langle\hat{L}_y\rangle^2} = \frac{\hbar}{\sqrt{2}}\sqrt{\ell(\ell+1) - m^2}.$$

Because $\langle\hat{L}_x\rangle = \langle\hat{L}_y\rangle = 0$ and $\langle\hat{L}_x^2\rangle = \langle\hat{L}_y^2\rangle = \frac{\hbar^2}{2}\big[\,\ell(\ell+1) - m^2\,\big]$, we have $\Delta L_x = \Delta L_y$.

(3) (3 points) The product of the above-calculated uncertainties in $\hat{L}_x$ and $\hat{L}_y$ is given by

$$\Delta L_x \Delta L_y = \frac{\hbar^2}{2}\big[\,\ell(\ell+1) - m^2\,\big].$$

From the generalized uncertainty relation $\Delta A\Delta B \geq \frac{1}{2}\big|\langle[\hat{A},\hat{B}]\rangle\big|$ with $\hat{A} = \hat{L}_x$ and $\hat{A} = \hat{L}_y$, we have

$$\Delta L_x \Delta L_y \geq \frac{1}{2}\big|\langle[\hat{L}_x,\hat{L}_y]\rangle\big| = \frac{\hbar}{2}\big|\langle\ell m|\hat{L}_z|\ell m\rangle\big| = \frac{\hbar^2}{2}|m|.$$

Because $|m| \leq \ell$, we have $\ell(\ell+1) - m^2 \geq \ell(\ell+1) - \ell^2 = \ell \geq |m|$. Thus, the uncertainty relation is satisfied. When $m = \pm\ell$, the equality in the uncertainty relation holds.

4. (15 points) The spin of a spin-1/2 particle is in the state $|\psi\rangle = \cos\big(\theta/2\big)e^{-i\phi/2}\,|+\rangle + \sin\big(\theta/2\big)e^{i\phi/2}\,|-\rangle$. The density operator for this spin state is given by $\hat{\rho} = |\psi\rangle\langle\psi|$.

(1) (6 points) Find the density matrix $\rho(\theta,\phi)$ in the $\{|+\rangle, |-\rangle\}$ basis.

(2) (3 points) Show that $\big[\rho(\theta,\phi)\big]^2 = \rho(\theta,\phi)$.

(3) (3 points) Evaluate $\langle\hat{S}_z\rangle$ using $\rho(\theta,\phi)$.

(4) (3 points) Introduce $\hat{S}_\perp = \hat{S}_x\cos\phi + \hat{S}_y\sin\phi$. Evaluate $\langle\hat{S}_\perp\rangle$ using $\rho(\theta,\phi)$.

Formulas provided: $\langle\hat{A}\rangle = \mathrm{Tr}\big(\hat{\rho}\hat{A}\big) = \mathrm{Tr}\big(\rho A\big)$, $S_x = \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$, $S_y = \frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}$, $S_z = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$.

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(1) (6 points) We first find the matrix elements of $\hat{\rho}$ in the $\{|+\rangle, |-\rangle\}$ basis. For the matrix element $\langle+|\hat{\rho}|+\rangle$, we have

$$\begin{aligned}
\langle+|\hat{\rho}|+\rangle &= \langle+|\psi\rangle\langle\psi|+\rangle \\
&= \langle+|\big[\cos\big(\theta/2\big)e^{-i\phi/2}\,|+\rangle + \sin\big(\theta/2\big)e^{i\phi/2}\,|-\rangle\,\big]\big[\cos\big(\theta/2\big)e^{i\phi/2}\,\langle+| + \sin\big(\theta/2\big)e^{-i\phi/2}\,\langle-|\,\big]\,|+\rangle \\
&= \cos\big(\theta/2\big)e^{-i\phi/2}\cdot\cos\big(\theta/2\big)e^{i\phi/2} \\
&= \cos^2\big(\theta/2\big).
\end{aligned}$$

For the matrix element $\langle +|\hat{\rho}|-\rangle$, we have

$$\begin{aligned}\langle +|\hat{\rho}|-\rangle &= \langle +|\psi\rangle\langle\psi|-\rangle \\ &= \langle +|\big[\cos\big(\theta/2\big)e^{-i\phi/2}\,|+\rangle + \sin\big(\theta/2\big)e^{i\phi/2}\,|-\rangle\,\big]\big[\cos\big(\theta/2\big)e^{i\phi/2}\,\langle +| + \sin\big(\theta/2\big)e^{-i\phi/2}\,\langle -|\,\big]\,|-\rangle \\ &= \cos\big(\theta/2\big)e^{-i\phi/2}\cdot\sin\big(\theta/2\big)e^{-i\phi/2} \\ &= \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi}.\end{aligned}$$

For the matrix element $\langle -|\hat{\rho}|+\rangle$, we have

$$\begin{aligned}\langle -|\hat{\rho}|+\rangle &= \langle -|\psi\rangle\langle\psi|+\rangle \\ &= \langle -|\big[\cos\big(\theta/2\big)e^{-i\phi/2}\,|+\rangle + \sin\big(\theta/2\big)e^{i\phi/2}\,|-\rangle\,\big]\big[\cos\big(\theta/2\big)e^{i\phi/2}\,\langle +| + \sin\big(\theta/2\big)e^{-i\phi/2}\,\langle -|\,\big]\,|+\rangle \\ &= \sin\big(\theta/2\big)e^{i\phi/2}\cdot\cos\big(\theta/2\big)e^{i\phi/2} \\ &= \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi}.\end{aligned}$$

For the matrix element $\langle -|\hat{\rho}|-\rangle$, we have

$$\begin{aligned}\langle -|\hat{\rho}|-\rangle &= \langle -|\psi\rangle\langle\psi|-\rangle \\ &= \langle -|\big[\cos\big(\theta/2\big)e^{-i\phi/2}\,|+\rangle + \sin\big(\theta/2\big)e^{i\phi/2}\,|-\rangle\,\big]\big[\cos\big(\theta/2\big)e^{i\phi/2}\,\langle +| + \sin\big(\theta/2\big)e^{-i\phi/2}\,\langle -|\,\big]\,|-\rangle \\ &= \sin\big(\theta/2\big)e^{i\phi/2}\cdot\sin\big(\theta/2\big)e^{-i\phi/2} \\ &= \sin^2\big(\theta/2\big).\end{aligned}$$

Thus, the density matrix $\rho(\theta,\phi)$ in the $\{|+\rangle, |-\rangle\}$ basis is given by

$$\rho(\theta,\phi) = \begin{pmatrix} \cos^2\big(\theta/2\big) & \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi} \\ \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi} & \sin^2\big(\theta/2\big)\end{pmatrix}.$$

(2) (3 points) Squaring $\rho(\theta,\phi)$ yields

$$\begin{aligned}\big[\rho(\theta,\phi)\big]^2 &= \begin{pmatrix} \cos^2\big(\theta/2\big) & \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi} \\ \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi} & \sin^2\big(\theta/2\big)\end{pmatrix}\begin{pmatrix} \cos^2\big(\theta/2\big) & \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi} \\ \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi} & \sin^2\big(\theta/2\big)\end{pmatrix} \\ &= \begin{pmatrix} \cos^4\big(\theta/2\big) + \sin^2\big(\theta/2\big)\cos^2\big(\theta/2\big) & \sin\big(\theta/2\big)\cos^3\big(\theta/2\big)e^{-i\phi} + \sin^3\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi} \\ \sin\big(\theta/2\big)\cos^3\big(\theta/2\big)e^{-i\phi} + \sin^3\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi} & \sin^2\big(\theta/2\big)\cos^2\big(\theta/2\big) + \sin^4\big(\theta/2\big)\end{pmatrix} \\ &= \begin{pmatrix} \cos^2\big(\theta/2\big) & \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi} \\ \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi} & \sin^2\big(\theta/2\big) \qquad = \rho(\theta,\phi).\end{pmatrix}\end{aligned}$$

Thus, $\big[\rho(\theta,\phi)\big]^2 = \rho(\theta,\phi)$.

(3) (3 points) Evaluating $\langle \hat{S}_z\rangle$, we have

$$\begin{aligned}\mathrm{Tr}\,\langle \hat{S}_z\rangle &= \mathrm{Tr}\big(\hat{\rho}\hat{S}_z\big) = \mathrm{Tr}\big[\,\rho(\theta,\phi)S_z\,\big] \\ &= \frac{\hbar}{2}\,\mathrm{Tr}\left[\begin{pmatrix} \cos^2\big(\theta/2\big) & \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi} \\ \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi} & \sin^2\big(\theta/2\big)\end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1\end{pmatrix}\right] \\ &= \frac{\hbar}{2}\,\mathrm{Tr}\begin{pmatrix} \cos^2\big(\theta/2\big) & -\sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi} \\ \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi} & -\sin^2\big(\theta/2\big)\end{pmatrix} \\ &= \frac{\hbar}{2}\big[\,\cos^2\big(\theta/2\big) - \sin^2\big(\theta/2\big)\,\big] = \frac{\hbar}{2}\cos\theta.\end{aligned}$$

(4) (3 points) The representation matrix of $\hat{S}_\perp$ is given by

$$S_\perp = S_x\cos\phi + S_y\sin\phi = \frac{\hbar}{2}\begin{pmatrix} 0 & e^{-i\phi} \\ e^{i\phi} & 0\end{pmatrix}.$$

Evaluating $\langle \hat{S}_\perp \rangle$, we have

$$\begin{aligned}\langle \hat{S}_\perp \rangle &= \mathrm{Tr}\big(\hat{\rho}\hat{S}_\perp\big) = \mathrm{Tr}\big[\,\rho(\theta,\phi)S_\perp\,\big] \\ &= \frac{\hbar}{2}\,\mathrm{Tr}\left[\begin{pmatrix} \cos^2\big(\theta/2\big) & \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{-i\phi} \\ \sin\big(\theta/2\big)\cos\big(\theta/2\big)e^{i\phi} & \sin^2\big(\theta/2\big)\end{pmatrix}\begin{pmatrix} 0 & e^{-i\phi} \\ e^{i\phi} & 0\end{pmatrix}\right] \\ &= \frac{\hbar}{2}\,\mathrm{Tr}\begin{pmatrix} \sin\big(\theta/2\big)\cos\big(\theta/2\big) & \cos^2\big(\theta/2\big)e^{-i\phi} \\ \sin^2\big(\theta/2\big)e^{i\phi} & \sin\big(\theta/2\big)\cos\big(\theta/2\big)\end{pmatrix} = \frac{\hbar}{2}\sin\theta.\end{aligned}$$

5. (15 points) Consider the states of an electron in a linear three-atom molecule with equally spaced atoms 1, 2, and 3. The distance between any two neighboring atoms is d. Let $|\psi_1\rangle$, $|\psi_2\rangle$, and $|\psi_3\rangle$ denote the orthonormal eigenvectors of an observable $\hat{D}$ corresponding to an electron localized respectively in the vicinity of the atoms 1, 2, and 3. We have $\hat{D}\,|\psi_1\rangle = -d\,|\psi_1\rangle$, $\hat{D}\,|\psi_2\rangle = 0$, and $\hat{D}\,|\psi_3\rangle = d\,|\psi_3\rangle$. In the $\{|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle\}$ basis, the representation matrix H of the Hamiltonian operator $\hat{H}$ of the system is $H = \begin{pmatrix} E_0 & -V & 0 \\ -V & E_0 & -V \\ 0 & -V & E_0\end{pmatrix}$ with $V > 0$.

(1) (6 points) Find the eigenvalues and eigenvectors of $\hat{H}$. Express the eigenvectors of $\hat{H}$ in terms of $|\psi_1\rangle$, $|\psi_2\rangle$, and $|\psi_3\rangle$.

(2) (3 points) Suppose that the electron is in the ground state. What are the probabilities of finding the electron in the vicinity of the atoms 1, 2, and 3?

(3) (6 points) Suppose that the electron is in the state $|\psi_1\rangle$. What results can be obtained if the energy of the system is measured? Evaluate the average of the energy, $\langle \hat{H} \rangle$, and the uncertainty in the energy, ΔE.

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(1) (6 points) Let E be an eigenvalue of $\hat{H}$. Let $|\varphi\rangle = \alpha\,|\psi_1\rangle + \beta\,|\psi_2\rangle + \gamma\,|\psi_3\rangle$ be the corresponding eigenvector of $\hat{H}$. In the basis $\{|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle\}$, the eigenfunction of $\hat{H}$ with the eigenvalue E reads $\varphi = \begin{pmatrix}\alpha \\ \beta \\ \gamma\end{pmatrix}$. In the $\{|\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle\}$ basis, the eigenequation $\hat{H}\,|\varphi\rangle = E\,|\varphi\rangle$ of $\hat{H}$ reads

$$\begin{pmatrix} E_0 & -V & 0 \\ -V & E_0 & -V \\ 0 & -V & E_0\end{pmatrix}\begin{pmatrix}\alpha \\ \beta \\ \gamma\end{pmatrix} = E\begin{pmatrix}\alpha \\ \beta \\ \gamma\end{pmatrix}.$$

Evaluating the matrix product on the left hand side, we have

$$\begin{aligned} (E_0 - E)\alpha \;-\; & V\beta \;&&-\; 0\gamma \;&&=\; 0, \\ -V\alpha \;+\; & (E_0 - E)\beta \;-\; & V\gamma \;&=\; 0, \\ -\; & V\beta \;+\; & (E_0 - E)\gamma \;&=\; 0. \end{aligned}$$

The above equations are homogeneous linear algebraic equations for the coefficients α, β, and γ. From the necessary and sufficient condition for the existence of nontrivial solutions, we have the following secular equation

$$\det\begin{vmatrix} E_0 - E & -V & 0 \\ -V & E_0 - E & -V \\ 0 & -V & E_0 - E\end{vmatrix} = 0.$$

Evaluating the determinant on the left hand side of the above equation yields

$$(E_0 - E)^3 - 2V^2(E_0 - E) = 0$$

from which we obtain the following eigenvalues of $\hat{H}$

$$E_1 = E_0 - \sqrt{2}\,V,\; E_2 = E_0,\; E_0 + \sqrt{2}\,V.$$

The eigenvalues of $\hat{H}$ have been arranged in the increasing order. $E_1 = E_0 - \sqrt{2}\,V$ is the ground-state energy.

Finding the eigenvector of $\hat{H}$ with the eigenvalue E_1. To find the eigenvector of $\hat{H}$ with the eigenvalue $E_1 = E_0 - \sqrt{2}\,V$, we insert $E = E_1 = E_0 - \sqrt{2}\,V$ into the homogeneous linear algebraic equations for the coefficients α, β, and γ. We have

$$\begin{array}{rcrcrcl} \sqrt{2}\,\alpha & - & \beta & & & = & 0, \\ -\alpha & + & \sqrt{2}\,\beta & - & V\gamma & = & 0, \\ & - & \beta & + & \sqrt{2}\,\gamma & = & 0. \end{array}$$

Using the above equations, we obtain express β and γ in terms of α. We have

$$\begin{aligned} \beta &= \sqrt{2}\,\alpha, \\ \gamma &= \frac{1}{\sqrt{2}}\beta = \alpha. \end{aligned}$$

From the normalization condition $|\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1$, we have $4|\alpha|^2 = 1$ which yields $|\alpha| = \dfrac{1}{2}$. We choose $\alpha = \dfrac{1}{2}$. We then have $\beta = \dfrac{1}{\sqrt{2}}$ and $\gamma = \dfrac{1}{2}$. The eigenvector of $\hat{H}$ with the eigenvalue $E_1 = E_0 - \sqrt{2}\,V$ is then given by

$$|\varphi_1\rangle = \frac{1}{2}\big[\,|\psi_1\rangle + \sqrt{2}\,|\psi_2\rangle + |\psi_3\rangle\,\big].$$

Finding the eigenvector of $\hat{H}$ with the eigenvalue E_2. To find the eigenvector of $\hat{H}$ with the eigenvalue $E_2 = E_0$, we insert $E = E_2 = E_0$ into the homogeneous linear algebraic equations for the coefficients α, β, and γ. We have

$$\begin{array}{rcrcrcl} & - & \beta & & & = & 0, \\ -\alpha & & & - & \gamma & = & 0, \\ & - & \beta & & & = & 0. \end{array}$$

From the above equations, we obtain

$$\begin{aligned} \beta &= 0, \\ \gamma &= -\alpha. \end{aligned}$$

From the normalization condition $|\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1$, we have $2|\alpha|^2 = 1$ which yields $|\alpha| = \dfrac{1}{\sqrt{2}}$. We choose $\alpha = \dfrac{1}{\sqrt{2}}$. We then have $\gamma = -\dfrac{1}{\sqrt{2}}$. The eigenvector of $\hat{H}$ with the eigenvalue $E_2 = E_0$ is then given by

$$|\varphi_2\rangle = \frac{1}{\sqrt{2}}\big[\,|\psi_1\rangle - |\psi_3\rangle\,\big].$$

Finding the eigenvector of $\hat{H}$ with the eigenvalue E_3. To find the eigenvector of $\hat{H}$ with the eigenvalue $E_3 = E_0 + \sqrt{2}\,V$, we insert $E = E_3 = E_0 + \sqrt{2}\,V$ into the homogeneous linear algebraic equations for the coefficients α, β, and γ. We have

$$\begin{array}{rcrcrcl} -\sqrt{2}\,\alpha & - & \beta & & & = & 0, \\ -\alpha & - & \sqrt{2}\,\beta & - & \gamma & = & 0, \\ & - & \beta & - & \sqrt{2}\,\gamma & = & 0. \end{array}$$

Using the above equations, we obtain express β and γ in terms of α. We have

$$\begin{aligned} \beta &= -\sqrt{2}\,\alpha, \\ \gamma &= -\frac{1}{\sqrt{2}}\beta = \alpha. \end{aligned}$$

From the normalization condition $|\alpha|^2+|\beta|^2+|\gamma|^2=1$, we have $4|\alpha|^2=1$ which yields $|\alpha|=\frac{1}{2}$. We choose $\alpha=\frac{1}{2}$. We then have $\beta=-\frac{1}{\sqrt{2}}$ and $\gamma=\frac{1}{2}$. The eigenvector of $\hat{H}$ with the eigenvalue $E_3=E_0+\sqrt{2}\,V$ is then given by

$$|\varphi_3\rangle=\frac{1}{2}\big[\,|\psi_1\rangle-\sqrt{2}\,|\psi_2\rangle+|\psi_3\rangle\,\big].$$

(2) (3 points) Because the electron is in the ground state, the state vector is given by

$$|\psi_{gs}\rangle=|\varphi_1\rangle=\frac{1}{2}\big[\,|\psi_1\rangle+\sqrt{2}\,|\psi_2\rangle+|\psi_3\rangle\,\big].$$

The probability of finding the electron in the vicinity of atom 1 is given by

$$\mathscr{P}(1)=|\langle\psi_1|\psi_{gs}\rangle|^2=\frac{1}{4}\Big|\langle\psi_1|\big[\,|\psi_1\rangle+\sqrt{2}\,|\psi_2\rangle+|\psi_3\rangle\,\big]\rangle\Big|^2=\frac{1}{4}.$$

The probability of finding the electron in the vicinity of atom 2 is given by

$$\mathscr{P}(2)=|\langle\psi_2|\psi_{gs}\rangle|^2=\frac{1}{4}\Big|\langle\psi_2|\big[\,|\psi_1\rangle+\sqrt{2}\,|\psi_2\rangle+|\psi_3\rangle\,\big]\rangle\Big|^2=\frac{1}{2}.$$

The probability of finding the electron in the vicinity of atom 3 is given by

$$\mathscr{P}(3)=|\langle\psi_3|\psi_{gs}\rangle|^2=\frac{1}{4}\Big|\langle\psi_3|\big[\,|\psi_1\rangle+\sqrt{2}\,|\psi_2\rangle+|\psi_3\rangle\,\big]\rangle\Big|^2=\frac{1}{4}.$$

Note that $\mathscr{P}(1)+\mathscr{P}(2)+\mathscr{P}(3)=1$.

(3) (6 points) We know that the result of a measurement of the energy is an eigenvalue of $\hat{H}$. To find all the possible results for measuring the energy, we expand $|\psi_1\rangle$ in terms of the eigenvectors of $\hat{H}$. We have

$$|\psi_1\rangle=\frac{1}{2}\big[\,|\varphi_1\rangle+\sqrt{2}\,|\varphi_2\rangle+|\varphi_3\rangle\,\big].$$

Since all the three eigenvectors of $\hat{H}$ appear in the expansion of $|\psi_1\rangle$, the possible results for measuring the energy are $E_1=E_0-\sqrt{2}\,V$, $E_2=E_0$, and $E_3=E_0+\sqrt{2}\,V$. We can obtain the probabilities for obtaining these results using the above expansion of $|\psi_1\rangle$ in terms of the eigenvectors of $\hat{H}$. We have

$$\begin{aligned}\mathscr{P}_{\psi_1}(E_1)&=|\langle\varphi_1|\psi_1\rangle|^2=\frac{1}{4},\\ \mathscr{P}_{\psi_1}(E_2)&=|\langle\varphi_2|\psi_1\rangle|^2=\frac{1}{2},\\ \mathscr{P}_{\psi_1}(E_3)&=|\langle\varphi_3|\psi_1\rangle|^2=\frac{1}{4}.\end{aligned}$$

We list these probabilities together with the possible results in the following.

Result	Probability
$E_1=E_0-\sqrt{2}\,V$	$\mathscr{P}_{\psi_1}(E_1)=\frac{1}{4}$
$E_2=E_0$	$\mathscr{P}_{\psi_1}(E_2)=\frac{1}{2}$
$E_3=E_0+\sqrt{2}\,V$	$\mathscr{P}_{\psi_1}(E_3)=\frac{1}{4}$

Making use of the above results, we can easily evaluate the average of the energy as follows

$$\begin{aligned}\langle\hat{H}\rangle&=\mathscr{P}_{\psi_1}(E_1E_1+\mathscr{P}_{\psi_1}(E_2E_2+\mathscr{P}_{\psi_1}(E_3E_3\\ &=\frac{1}{4}\times(E_0-\sqrt{2}\,V)+\frac{1}{2}\times E_0+\frac{1}{4}\times(E_0+\sqrt{2}\,V)\\ &=E_0.\end{aligned}$$

To find the uncertainty of the energy, we evaluate the average of $\hat{H}^2$. We have

$$\begin{aligned}\langle \hat{H}^2 \rangle &= \mathscr{P}_{\psi_1}(E_1 E_1^2 + \mathscr{P}_{\psi_1}(E_2 E_2^2 + \mathscr{P}_{\psi_1}(E_3 E_3^2 \\ &= \frac{1}{4} \times (E_0 - \sqrt{2}\, V)^2 + \frac{1}{2} \times E_0^2 + \frac{1}{4} \times (E_0 + \sqrt{2}\, V)^2 \\ &= E_0^2 + V^2.\end{aligned}$$

The uncertainty in the energy is then given by

$$\Delta E = \sqrt{\langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2} = \sqrt{E_0^2 + V^2 - E_0^2} = V.$$

6. (15 points) The Hamiltonian operator of a system is given by $\hat{H}(t) = \hat{H}_0 + \hat{V}(t)$. The eigenvalues of the time-independent part $\hat{H}_0$ are $E_0/2$ and $-E_0/2$ with the corresponding normalized eigenvectors of $\hat{H}_0$ respectively given by $|+\rangle$ and $|-\rangle$. For the convenience of studying the time evolution of this system, we choose the initial time to be $t = 0$ and introduce $|\tilde{\psi}(t)\rangle = \hat{U}_1^\dagger |\psi(t)\rangle$ with $|\psi(t)\rangle$ the state vector of the system at time t in the Schrödinger picture.

(1) (5 points) The time evolution of $|\tilde{\psi}(t)\rangle$ is governed by $i\hbar \dfrac{d}{dt} |\tilde{\psi}(t)\rangle = \hat{H}_1 |\tilde{\psi}(t)\rangle$. Find $\hat{H}_1$.

(2) (5 points) In the $\{|+\rangle, |-\rangle\}$ basis, the representation matrices of $\hat{V}(t)$ and $\hat{U}_1$ are respectively given by $V(t) = \begin{pmatrix} 0 & V_0 e^{-i\omega_0 t} \\ V_0 e^{i\omega_0 t} & 0 \end{pmatrix}$ and $U_1(t) = \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix}$. Find the representation matrix $\tilde{U}(t)$ of $\hat{\tilde{U}}(t)$ in the $\{|+\rangle, |-\rangle\}$ basis such that $|\tilde{\psi}(t)\rangle = \hat{\tilde{U}}(t) |\tilde{\psi}(0)\rangle$.

(3) (5 points) Assume that the system is in the state $|-\rangle$ at $t = 0$. What is the probability $\mathscr{P}_{+-}(t)$ of finding the system in the state $|+\rangle$ at $t > 0$?

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(1) (5 points) Differentiating $|\tilde{\psi}(t)\rangle = \hat{U}_1^\dagger(t) |\psi(t)\rangle$ with respect to t and then multiplying both sides of the resultant equation with $i\hbar$, we obtain

$$i\hbar \frac{d}{dt} |\tilde{\psi}(t)\rangle = i\hbar \hat{U}_1^\dagger(t) \frac{d\,|\psi(t)\rangle}{dt} + i\hbar \frac{d\hat{U}_1^\dagger(t)}{dt} |\psi(t)\rangle\,.$$

Making use of $i\hbar \dfrac{d\,|\psi(t)\rangle}{dt} = \hat{H}(t) |\psi(t)\rangle$, we have

$$i\hbar \frac{d}{dt} |\tilde{\psi}(t)\rangle = \hat{U}_1^\dagger(t) \hat{H}(t) |\psi(t)\rangle + i\hbar \frac{d\hat{U}_1^\dagger(t)}{dt} |\psi(t)\rangle\,.$$

Making use of $\hat{U}_1(t)\hat{U}_1^\dagger(t) = 1$, we have

$$\begin{aligned} i\hbar \frac{d}{dt} |\tilde{\psi}(t)\rangle &= \hat{U}_1^\dagger(t) \hat{H}(t) \hat{U}_1(t) \hat{U}_1^\dagger(t) |\psi(t)\rangle + i\hbar \frac{d\hat{U}_1^\dagger(t)}{dt} \hat{U}_1(t) \hat{U}_1^\dagger(t) |\psi(t)\rangle \\ &= \hat{U}_1^\dagger(t) \hat{H}(t) \hat{U}_1(t) |\tilde{\psi}(t)\rangle + i\hbar \frac{d\hat{U}_1^\dagger(t)}{dt} \hat{U}_1(t) |\tilde{\psi}(t)\rangle\,. \end{aligned}$$

Differentiating $\hat{U}_1^\dagger(t)\hat{U}_1(t) = 1$ with respect to t, we have

$$\frac{d\hat{U}_1^\dagger(t)}{dt} \hat{U}_1(t) + \hat{U}_1^\dagger(t) \frac{d\hat{U}_1(t)}{dt} = 0.$$

We then have

$$\frac{d\hat{U}_1^\dagger(t)}{dt} \hat{U}_1(t) = -\hat{U}_1^\dagger(t) \frac{d\hat{U}_1(t)}{dt}.$$

Making use of the above result, we can rewrite the equation for $|\tilde{\psi}\rangle(t)$ as

$$\begin{aligned} i\hbar\frac{d}{dt}|\tilde{\psi}(t)\rangle &= \hat{U}_1^\dagger(t)\hat{H}(t)\hat{U}_1(t)|\tilde{\psi}(t)\rangle - i\hbar\hat{U}_1^\dagger(t)\frac{d\hat{U}_1(t)}{dt}|\tilde{\psi}(t)\rangle \\ &= \left[\hat{U}_1^\dagger(t)\hat{H}(t)\hat{U}_1(t) - i\hbar\hat{U}_1^\dagger(t)\frac{d\hat{U}_1(t)}{dt}\right]|\tilde{\psi}(t)\rangle . \end{aligned}$$

Comparing the above equation with $i\hbar\frac{d}{dt}|\tilde{\psi}(t)\rangle = \hat{H}_1|\tilde{\psi}(t)\rangle$, we obtain

$$\hat{H}_1 = \hat{U}_1^\dagger(t)\hat{H}(t)\hat{U}_1(t) - i\hbar\hat{U}_1^\dagger(t)\frac{d\hat{U}_1(t)}{dt}.$$

(2) (5 points) In the $\{|+\rangle, |-\rangle\}$ basis, the representation matrix of $\hat{H}(t)$ is given by

$$H(t) = \begin{pmatrix} E_0/2 & 0 \\ 0 & -E_0/2 \end{pmatrix} + \begin{pmatrix} 0 & V_0 e^{-i\omega_0 t} \\ V_0 e^{i\omega_0 t} & 0 \end{pmatrix} = \begin{pmatrix} E_0/2 & V_0 e^{-i\omega_0 t} \\ V_0 e^{i\omega_0 t} & -E_0/2 \end{pmatrix}.$$

In the $\{|+\rangle, |-\rangle\}$ basis, the representation matrix H_1 of $\hat{H}_1$ is given by

$$\begin{aligned} H_1 &= U_1^\dagger(t)H(t)U_1(t) - i\hbar U_1^\dagger(t)\frac{dU_1(t)}{dt} \\ &= \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix}^\dagger \left[\begin{pmatrix} E_0/2 & V_0 e^{-i\omega_0 t} \\ V_0 e^{i\omega_0 t} & -E_0/2 \end{pmatrix}\begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix} - i\hbar\frac{d}{dt}\begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix} \right] \\ &= \begin{pmatrix} e^{i\omega_0 t/2} & 0 \\ 0 & e^{-i\omega_0 t/2} \end{pmatrix} \left[\begin{pmatrix} (E_0/2)e^{-i\omega_0 t/2} & V_0 e^{-i\omega_0 t/2} \\ V_0 e^{i\omega_0 t/2} & -(E_0/2)e^{i\omega_0 t/2} \end{pmatrix} - \begin{pmatrix} (\hbar\omega_0/2)e^{-i\omega_0 t/2} & 0 \\ 0 & -(\hbar\omega_0/2)e^{i\omega_0 t/2} \end{pmatrix} \right] \\ &= \begin{pmatrix} e^{i\omega_0 t/2} & 0 \\ 0 & e^{-i\omega_0 t/2} \end{pmatrix}\begin{pmatrix} [(E_0 - \hbar\omega_0)/2]e^{-i\omega_0 t/2} & V_0 e^{-i\omega_0 t/2} \\ V_0 e^{i\omega_0 t/2} & -[(E_0 - \hbar\omega_0)/2]e^{i\omega_0 t/2} \end{pmatrix} \\ &= \begin{pmatrix} (E_0 - \hbar\omega_0)/2 & V_0 \\ V_0 & -(E_0 - \hbar\omega_0)/2 \end{pmatrix}. \end{aligned}$$

Since H_1 is independent on t, the representation matrix $\tilde{U}(t)$ of $\hat{\tilde{U}}(t)$ in the $\{|+\rangle, |-\rangle\}$ basis is given by

$$\tilde{U}(t) = e^{-iH_1 t/\hbar}.$$

Expanding the exponential function in the above equation yields

$$\tilde{U}(t) = \sum_{n=0}^{\infty} \frac{(-it/\hbar)^n}{n!} H_1^n.$$

We now find the powers of H_1. For H_1^2, we have

$$\begin{aligned} H_1^2 &= \begin{pmatrix} (E_0 - \hbar\omega_0)/2 & V_0 \\ V_0 & -(E_0 - \hbar\omega_0)/2 \end{pmatrix}\begin{pmatrix} (E_0 - \hbar\omega_0)/2 & V_0 \\ V_0 & -(E_0 - \hbar\omega_0)/2 \end{pmatrix} \\ &= \begin{pmatrix} (E_0 - \hbar\omega_0)^2/4 + V_0^2 & 0 \\ 0 & (E_0 - \hbar\omega_0)^2/4 + V_0^2 \end{pmatrix} \\ &= (\hbar\omega)^2 I, \end{aligned}$$

where $\hbar\omega = \sqrt{(E_0 - \hbar\omega_0)^2/4 + V_0^2}$ and I is the 2×2 unit matrix,

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

For H_1^n, we have

$$H_1^n = \begin{cases} (\hbar\omega)^{2j} I, & n = 2j,\ j = 0, 1, 2, \cdots, \\ (\hbar\omega)^{2j} H_1, & n = 2j+1,\ j = 0, 1, 2, \cdots. \end{cases}$$

Separating the even and odd powers in H_1 in the expansion of $\tilde{U}(t)$, we have

$$\begin{aligned} \tilde{U}(t) &= \sum_{j=0}^{\infty} \frac{(-it/\hbar)^{2j}}{(2j)!} H_1^{2j} + \sum_{j=0}^{\infty} \frac{(-it/\hbar)^{2j+1}}{(2j+1)!} H_1^{2j+1} \\ &= I \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j)!} (\omega t)^{2j} - \frac{i}{\hbar\omega} H_1 \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)!} (\omega t)^{2j+1} \\ &= I \cos(\omega t) - \frac{i}{\hbar\omega} H_1 \sin(\omega t) \\ &= \begin{pmatrix} \cos(\omega t) - i[(E_0 - \hbar\omega_0)/2\hbar\omega] \sin(\omega t) & -i(V_0/\hbar\omega) \sin(\omega t) \\ -i(V_0/\hbar\omega) \sin(\omega t) & \cos(\omega t) + i[(E_0 - \hbar\omega_0)/2\hbar\omega] \sin(\omega t) \end{pmatrix}. \end{aligned}$$

(3) (5 points) The probability $\mathscr{P}_{+-}(t)$ of finding the system in the state $|+\rangle$ at $t > 0$ is given by

$$\mathscr{P}_{+-}(t) = \left| \langle + | \psi(t) \rangle \right|^2.$$

Making use of $|\psi(t)\rangle = \hat{U}_1 |\tilde{\psi}(t)\rangle$, we have

$$\mathscr{P}_{+-}(t) = \left| \langle + | \hat{U}_1(t) | \tilde{\psi}(t) \rangle \right|^2.$$

Making use of $|\tilde{\psi}(t)\rangle = \hat{\tilde{U}}(t) |\tilde{\psi}(0)\rangle$, we have

$$\mathscr{P}_{+-}(t) = \left| \langle + | \hat{U}_1(t) \hat{\tilde{U}}(t) | \tilde{\psi}(0) \rangle \right|^2.$$

Making use of $|\tilde{\psi}(0)\rangle = |-\rangle$, we have

$$\mathscr{P}_{+-}(t) = \left| \langle + | \hat{U}_1(t) \hat{\tilde{U}}(t) | - \rangle \right|^2.$$

Making use of the representation matrices of $\hat{U}_1(t)$ and $\hat{\tilde{U}}(t)$, we have

$$\begin{aligned} U_1(t)\tilde{U}(t) &= \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix} \begin{pmatrix} \cos(\omega t) - i[(E_0 - \hbar\omega_0)/2\hbar\omega] \sin(\omega t) & -i(V_0/\hbar\omega) \sin(\omega t) \\ -i(V_0/\hbar\omega) \sin(\omega t) & \cos(\omega t) + i[(E_0 - \hbar\omega_0)/2\hbar\omega] \sin(\omega t) \end{pmatrix} \\ &= \begin{pmatrix} \{\cos(\omega t) - i[(E_0 - \hbar\omega_0)/2\hbar\omega] \sin(\omega t)\} e^{-i\omega_0 t/2} & -i(V_0/\hbar\omega) \sin(\omega t) e^{-i\omega_0 t/2} \\ -i(V_0/\hbar\omega) \sin(\omega t) e^{i\omega_0 t/2} & \{\cos(\omega t) + i[(E_0 - \hbar\omega_0)/2\hbar\omega] \sin(\omega t)\} e^{i\omega_0 t/2} \end{pmatrix}. \end{aligned}$$

$\mathscr{P}_{+-}(t)$ is then given by

$$\begin{aligned} \mathscr{P}_{+-}(t) &= \left| \begin{pmatrix} 1 & 0 \end{pmatrix} U_1(t)\tilde{U}(t) \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right|^2 \\ &= \left| \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} -i(V_0/\hbar\omega) \sin(\omega t) e^{-i\omega_0 t/2} \\ \{\cos(\omega t) + i[(E_0 - \hbar\omega_0)/2\hbar\omega] \sin(\omega t)\} e^{i\omega_0 t/2} \end{pmatrix} \right|^2 \\ &= \left(\frac{V_0}{\hbar\omega} \right)^2 \sin^2(\omega t). \end{aligned}$$